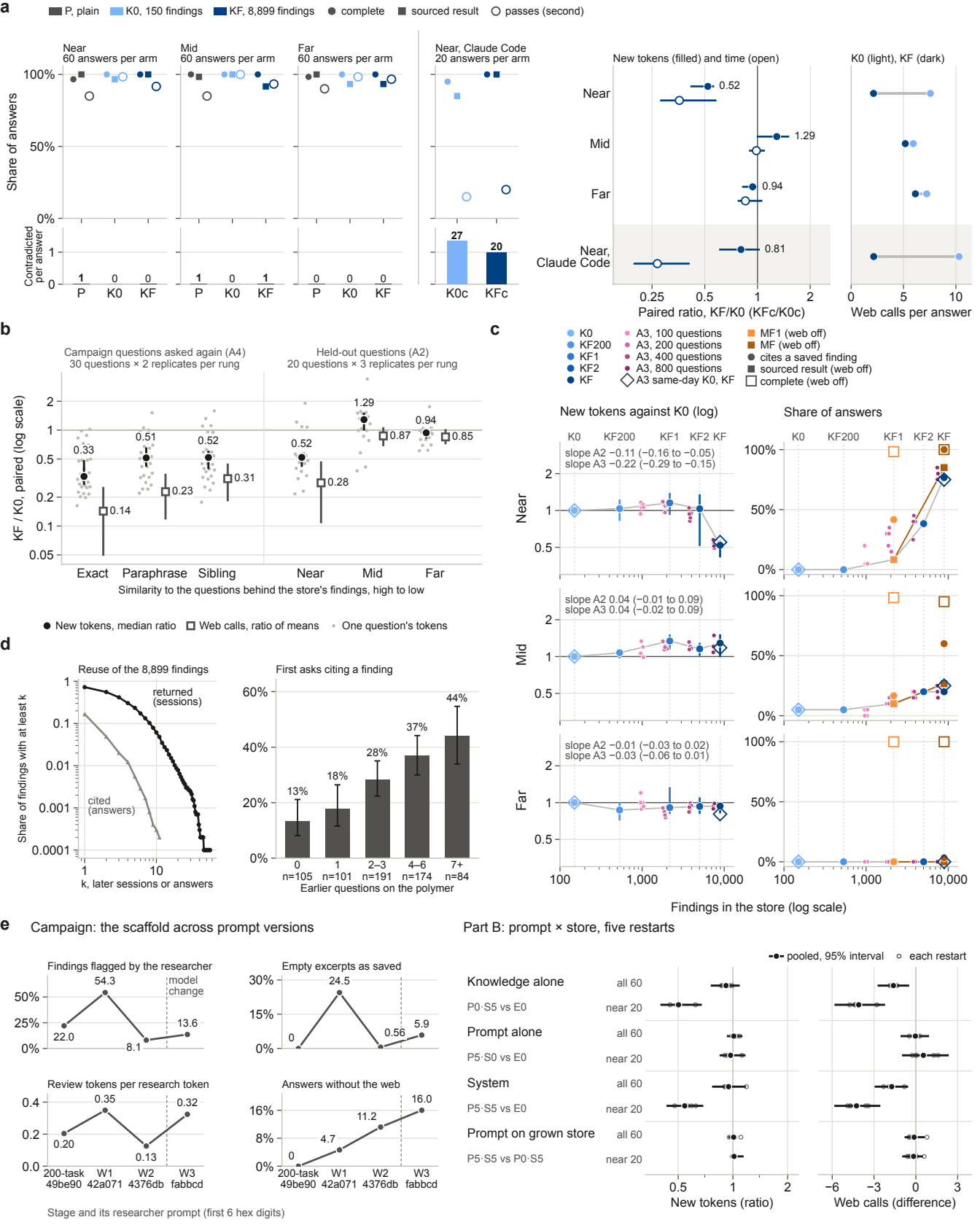

a
P, plain
K0, 150 findings
KF, 8,899 findings
complete
sourced result
passes (second)
Near
60 answers per arm
Mid
60 answers per arm
Far
60 answers per arm
Near, Claude Code
20 answers per arm
Share of answers
100%
50%
0%
Contradicted per answer
1
0
P
K0
KF
K0c
KFc
27
20
New tokens (filled) and time (open)
K0 (light), KF (dark)
Near
Mid
Far
Near, Claude Code
0.52
1.29
0.94
0.81
0.25
0.5
1
2
Paired ratio, KF/K0 (KFc/K0c)
0
5
10
Web calls per answer
b
Campaign questions asked again (A4)
30 questions × 2 replicates per rung
Held-out questions (A2)
20 questions × 3 replicates per rung
KF / K0, paired (log scale)
2
1
0.5
0.2
0.1
0.05
0.33
0.14
0.51
0.23
0.52
0.31
0.52
0.28
1.29
0.87
0.94
0.85
Exact
Paraphrase
Sibling
Near
Mid
Far
Similarity to the questions behind the store's findings, high to low
New tokens, median ratio
Web calls, ratio of means
One question's tokens
c
K0
KF200
KF1
KF2
KF
A3, 100 questions
A3, 200 questions
A3, 400 questions
A3, 800 questions
A3 same-day K0, KF
MF1 (web off)
MF (web off)
cites a saved finding
sourced result (web off)
complete (web off)
New tokens against K0 (log)
Share of answers
K0
KF200
KF1
KF2
KF
Near
slope A2 −0.11 (−0.16 to −0.05)
slope A3 −0.22 (−0.29 to −0.15)
Mid
slope A2 0.04 (−0.01 to 0.09)
slope A3 0.04 (−0.02 to 0.09)
Far
slope A2 −0.01 (−0.03 to 0.02)
slope A3 −0.03 (−0.06 to 0.01)
2
1
0.5
100%
50%
0%
100
1,000
10,000
Findings in the store (log scale)
d
Reuse of the 8,899 findings
Share of findings with at least k
1
0.1
0.01
0.001
0.0001
returned (sessions)
cited (answers)
1
10
k, later sessions or answers
First asks citing a finding
60%
40%
20%
0%
13%
18%
28%
37%
44%
0
n=105
1
n=101
2–3
n=191
4–6
n=174
7+
n=84
Earlier questions on the polymer
e
Campaign: the scaffold across prompt versions
Findings flagged by the researcher
50%
25%
0%
54.3
22.0
8.1
13.6
model change
Empty excerpts as saved
30%
15%
0%
24.5
0
0.56
5.9
Review tokens per research token
0.4
0.2
0.0
0.35
0.20
0.13
0.32
Answers without the web
16%
8%
0%
16.0
11.2
4.7
0
200-task 49be90
W1 42a071
W2 4376db
W3 fabbcd
Stage and its researcher prompt (first 6 hex digits)
Part B: prompt × store, five restarts
pooled, 95% interval
each restart
Knowledge alone
P0·S5 vs E0
Prompt alone
P5·S0 vs E0
System
P5·S5 vs E0
Prompt on grown store
P5·S5 vs P0·S5
all 60
near 20
0.5
1
2
New tokens (ratio)
−6
−3
0
3
Web calls (difference)